# Smart & sustainable agriculture to reduce irrigation water waste, improve resource efficiency & fight desert locusts

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***Abstract*—Sustainable agriculture is now a top global priority due to the world's population expansion, environmental deterioration, and resource shortages, climate change. Today, artificial intelligence, a true driving force behind sustainable agriculture, offers creative solutions to optimize the use of resources in order to improve agricultural yields, reduce risks and promote climate resilience. This research proposes a robust multimodal deep learning framework, combining Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, to reduce irrigation water waste, enhance resource efficiency, fight the Algerian desert locust, and improve productivity while promoting sustainability and inclusiveness. This framework integrates IoT signals, and public agriculture knowledge & skills while embedding environmental education to reduce misinformation and promote sustainability and inclusiveness. Methodological rigor is ensured via detailed preprocessing, justified hyperparameters, cross-validation, error decomposition, and regularization strategies. Statistical validity is demonstrated with confidence intervals, p-values, and multiple-run variability, and reproducibility is ensured through publicly available code, detailed hyperparameter ranges, and hardware specifications.**

***Index Terms*—Agricultural Environment Monitoring, Smart and sustainable farming, Environmental Education, Fighting Algerian desert locust, Multimodal deep learning, Reducing irrigation water waste, Sensors (IoT)**

## A list of acronyms that includes nomenclature and units

- AI: Artificial Intelligence
- ALE: Automated Learning Environment
- API: Interface of Application Programming
- CNN: Convolutional NN
- CSA: Climate-Smart Agriculture
- DL: Deep Learning
- FFNN: Feed-Forward Neural Network
- GMT: Greenwich Mean Time
- LSTM: Memory with Long Short-Term
- MAE: Mean of Absolute Error
- MSA: Multi-head Self-Attention
- MSE: Mean of Square Error
- NB: Naive Bayes
- NN: Network of Neurons
- PCT: Positional Convolutional Transform
- RMSE: Root-Mean of Square Error
- RNN: Recurrent NN
- RF: Random Forest
- SVM: Machine with Support Vector
- WSN: Wireless Sensor Network

## Highlights

- The study relies on multimodal DL (social media, IoT & smart data) to gather information
- It seeks to reduce irrigation water waste, fight desert locust & improve productivity
- Smart Farming relies on Multimodal DL to reduce water waste and fight desert locust
- Smart farming concludes with advances to improve productivity & promote sustainability
- It aims to reduce water waste, fight desert locust, improve productivity, promote sustainability

## Graphical abstract

Fig. 1. Graphical abstract

## I. Introduction

Agriculture, exacerbated by climate change, faces increasing challenges related to the efficient management and optimization of water resources for agricultural productivity [18]. Developing and implementing prevention strategies must include strengthening human and material resources to control the desert locust [8] (Schistocerca gregaria) in Africa and Asia. Emerging technologies must be integrated into the strategies to establish effective, cost-effective, and environmentally friendly locust control systems [8] at the national and regional levels. Frequent infestations of the destructive desert locust [27], feeding on any edible substance in its path and able to devour crops its own weight, have resulted in enormous environmental and economic losses. This study develops a real-time smart irrigation system, enhancing resource efficiency and fighting desert locusts [8] using emerging technology (DL & IoT). This system enables an efficient water management, locust control and supporting sustainable agriculture in North Africa (Algeria) to improve agricultural productivity.

The objective of this study is to reduce irrigation water waste and control desert locusts [8] in Algeria. This study contributes in six main ways:

1) Modeling of the new deep learning-based multimodal hybrid model for smart and sustainable agriculture using CNN-LSTM;
2) Creating a hybrid CNN-LSTM model based on low-level content learning from sensors (IoT) to automatically and efficiently capture information about soil moisture and dryness in real time while verifying the veracity of the information using knowledge from Smart Data and also information from social media;
3) Enabling the collection of relevant and environmental information on the presence of desert locusts [8] in Algeria, and verify it, via sensors (IoT), smart data or social networks;
4) Overcoming the shortcomings of previous work and surpassing them, the event-agnostic model filters content from sensors, smart data, and social networks for upcoming events;
5) Given the damage already caused or that which would be caused by global warming with its extreme weather events, shortages of drinking water and drought, there is still a case to be made for devoting time and resources to climate change mitigation through artificial intelligence in smart and sustainable agriculture, improving agricultural productivity, and sustainability and inclusiveness.

This is how the rest of the paper is organized. Related work is presented in the next part, which is followed by the background on sustainable agriculture, and environmental education. The new Smart and Sustainable Farming Model to reduce irrigation water waste, enhance resource efficiency and fight desert locusts while improving agricultural productivity is presented in the fourth section, along with its design and implementation using sensors (IoT), smart data and social media through environmental education. All of the outcomes are discussed. We wrap up by discussing some potential future developments.

## II. Related Works

Managing water resources for agricultural productivity [18] was improved through developing a smart irrigation system using advanced technologies namely the Internet of the artificial intelligence (AI) and the Internet of Things (IoT). Frequent infestations of the destructive desert locust [8], [27], feeding on any edible substance in its path and able to devour crops its own weight, have resulted in enormous environmental and economic losses. Developing and implementing prevention strategies must include strengthening human and material resources to control the desert locust (Schistocerca gregaria) [8] in Africa and Asia. Emerging technologies must be integrated into the strategies to establish effective, cost-effective, and environmentally friendly locust control systems [8] at the national and regional levels. An innovative smart farming system [28] is presented, integrating Internet of Things (IoT) technologies, predictive algorithms, and automated control mechanisms to optimize irrigation and improve resource efficiency. A critical aspect of agriculture was also addressed, namely integrating IoT, Big Data and Artificial Intelligence technologies [19] to monitor and mitigate agricultural carbon emissions. The interest of integrated IoT smart sensors [29] for monitoring environmental factors namely humidity, temperature and soil composition, essential for crop growth, was satisfactorily demonstrated, as well as measuring greenhouse gases (carbon dioxide and methane) and nitrogen content in the soil with the aim of optimize the amount of fertilizers and pesticides to be used. L'eau et des est macronutriments sont apparemment régulés par le criquet pèlerin australien, Chortoicetes terminifera [9].

## III. Background on Sustainable Agriculture and Environmental education

### A. *Sustainable farming for Reducing Irrigation Water Waste & Fighting desert Locusts in Algeria*

The similarity is used in the environment of agriculture to gather event-related content such as:

- Very dry soil
- Dry soil

To remotely open irrigation valves;

- Wet soil
- Too wet soil

To remotely close valves.

Less formal content is brief, unstructured, and vague [1], [10], contains words from several languages, contains numerous spelling and grammatical errors, and varies considerably in length. In contrast, sensor signals (IoT) are generally accurate, once processed, cleaned, and compared with agricultural knowledge derived from smart data, such as photos of destructive desert locusts [8]. This type of information is used to trigger the remote alert system to combat these destructive locusts [8]: given their enormous numbers, the fog formed by all these locusts leaves their agricultural soil well cleaned.

- Fighting desert locust [8] in Algeria The locust alert consists of:
  - flying drones over them while
    - spraying them with insecticide
    - spraying them with harmful seeds
    - drowning them in background noise to scare them away
  - remotely opening valves to project harmful ducts to kill them
  - etc.

- Reducing resource waste
  - IoT sensors: IoT sensors, including soil moisture sensors, nutrient level monitors, and weather trackers, enable precise resource allocation by continuously monitoring conditions. For example, soil moisture sensors ensure irrigation is applied only when needed, thereby reducing water waste [18], [19].
  - Automated irrigation: Smart irrigation systems, driven by real-time data from IoT sensors, adjust water supply based on current conditions. This reduces over-irrigation, which can lead to resource waste and environmental problems.
  - Nutrient management: IoT systems track soil nutrient levels and provide data to guide farmers in applying fertilizer only where needed. This reduces the overuse of chemicals and minimizes resource waste.

They are frequently employed to express feelings like surprise, anxiety, and joy. Through APIs, a lot of networking platforms offer content access. Ingestion of content, removal of uninformative data, corpus learning of tagged content to ensure relevancy, and analysis of outcomes for situational awareness, alarms, and training are all done through online listening.

Using smart data and different social networks, the study suggests a novel CNN-LSTM-based transport monitoring, alerting, and education paradigm. RNN [7] and LSTM [5] are extended. This methodology integrates encapsulation from numerous social media sources and smart data, combining training with safety instruction, awareness, and alerts.

The first to simultaneously monitor Facebook and Twitter (many sources) to notify people of happenings was our earlier model, [1]. After introducing safety education, the intelligent [10] interface compiles pertinent content from many social media platforms and smart data. To increase awareness, the RNN-driven alerting paradigm [7] is employed. RNN training is improved by this model [7]. Last but not least, a deep CNN-LSTM hybrid model is constructed [3], [4], [7]. This paradigm effectively enhances training, alerting, and awareness. The models [5], [7] made use of several social media content sources (the entire web) and smart data.

### B. Background on Recent AI Models

CNN is a type of DL algorithm and a key technology in the modern field of AI. It consists of multiple convolutional layers with nonlinear activation functions, pooling layers, and fully connected layers, enabling them to capture complex patterns and textures in images, making them particularly well-suited for visual data interpretation applications. As a result, CNN plays a crucial role in medical image analysis, extracting and improving the accuracy and efficiency of image-based medical applications. While other techniques have been used, such as RNNs, LSTMs, or Bidirectional LSTMs, CNNs were the dominant approach in image processing until very recently. Transformers, on the other hand, initially developed for NLP, revolutionized the field of DL thanks to their unique architecture based on self-attention mechanisms. Visual transformers (ViTs) use this powerful concept to process images [24].

Explainable and hybrid AI models combine different AI techniques to achieve both transparency and high performance. They merge the predictive power of complex ”black box” models (such as deep learning) with the interpretability of simpler ”white box” models (such as rule-based systems). The goal is to leverage the accuracy of one type of model while using the other to explain the decision-making process, thus making the overall system more reliable and understandable [25].

### C. Background on Educational awareness

The purpose of this study is to illustrate the value of road safety education as well as the effects of various approaches on clean and efficient energy in terms of inclusivity and sustainability. Individual empowerment is essential to promoting clean and efficient energy use, especially through road safety education. An operational, practical, and economical instrument for managing transportation is examining the role of safety education in sustainability and inclusivity, renewable energy, and climate change adaptation [7]. Education about transport safety is crucial, particularly for the most vulnerable. Education can be done in a variety of ways, and none is superior. However, those with more education are better equipped to defend both themselves and others (see Table 3). Online information flows, including important content that can support prompt, efficient decision-making to enhance the traffic environment. It moves across many social media platforms and intelligent data. Different processing techniques, such as computational linguistics, statistical analysis, comparable document-aligned data, natural language processing, and neural learning, are developed for different purposes even though resources like tagged vocabulary and standardized data are available [1], [10].

TABLE I
COMPARING STRATEGIES AND TACTICS IN MODELS OF INCLUSIVENESS AND SUSTAINABILITY

| Models | Methods |
|---|---|
| **[12]** | Modeling the monitoring of the economic environment to support economic operations in sustainability and inclusiveness by combining deep learning with economic education from multiple sources |
| **[6]** | Alertness, Knowledge, Evaluation, and Instruction in Emergency Management and Sustainability with COVID-19 |
| **[3]** | Handling crises with safe information by raising awareness of inclusiveness, inclusivity, and education |
| **[18], [19]** | A comprehensive analysis of the combination of IoT, Big Data and AI to monitor carbon footprint and support broader sustainability goals in agriculture |
| **[20]** | Integrating advanced technologies: high-performance sensors and IoT could increase agricultural production and minimize economic losses |
| **Our New Approach** | CNN-LSTM-based Hybrid Model for Agricultural Environment monitoring via Environmental Education and Reliable Information, sustainability and inclusiveness |

The project intends to create methods for communities to effectively manage climate change through sustainable mobility by researching climate change and its effects on community vulnerability. It is theorized that by increasing human ability and knowledge, mainly through safety education, communities can create an effective long-term defense against collision and climatic threats. Skills, knowledge, and risk awareness could be directly impacted, while attitudes—both positive and negative—as well as poverty reduction, health, and information and resource access could be indirectly impacted. When it comes to planning for, recovering from, and responding to natural disasters, educated families, people, and communities should be the most adaptable and independent. Our new inclusive and sustainable strategy is contrasted with other methods in Table 2. The significance and effects of various safety education techniques are emphasized in this study. Sustainable transportation must be promoted in schools. An operational, practical, and economical instrument in traffic management is the examination of safety education's role in climate change adaptation, renewable energy, and sustainable traffic [14]. The public, especially vulnerable populations, should be made aware of the dangers posed by traffic. While there are various approaches to education, none is superior. People with education are able to defend themselves. By researching how climate change affects people's susceptibility, it seeks to create plans for sustainable travel that will assist communities in adjusting. The fundamental premise being investigated is that communities may effectively build long-term resilience by enhancing human ability and knowledge, especially through safety education. They could have an instant impact on knowledge and risk awareness. In terms of strategies and methods for capturing internet content, Table 4 contrasts our novel approach with those of existing models. The study's suggested hybrid real-time CNN-LSTM-based model to reduce irrigation water waste and fight desert locusts [8] in Algeria works best for evaluation and situational awareness (see Table 1). With training capabilities for effective and automated situation capture in real-time reports during significant events using tagged content and keywords/hashtags, the model leverages smart data, Sensor (IoT) information, and many social media sources to capture relevant information (see Table 3) from the agricultural environment.

TABLE II
COMPARING APPROACHES AND STRATEGIES IN EDUCATIONAL MODELS

| Models | Methods |
|---|---|
| **[13]** | An abstractive-extractive method for summarizing situational tweets during emergencies |
| **[7]** | Education-based crisis management using CNN-LSTM |
| **[14], [15]** | Higher education-related educational goals |
| **[16]** | The value of education in times of crisis |
| **[17]** | Education programs' opportunities and challenges |
| **[23]** | Digital Technologies in Education: Effects and Contributing Elements to Digital Transformation and Capacity |
| **[3]** | Using education to manage emergencies: COVID-19 Case Study |
| **[10]** | FFNN-based model with smart education |
| **[5]** | LSTM-based model with smart education |
| **[3]** | Use safe information to manage crises via awareness-raising & education |
| **Our New Approach** | CNN-LSTM-based Hybrid Model for Agricultural Environment monitoring via Environmental Education and Reliable Information, sustainability and inclusiveness |

TABLE III
CONTRASTING APPROACHES AND STRATEGIES IN MODELS TO RECORD WEB CONTENT

| Ref | Method | Sources of Capturing Content |
|---|---|---|
| **[17]** | Using games to teach during flood emergencies | Twitter |
| **[16]** | One of higher education's learning goals, particularly in relation to | Twitter |
| **[13]** | Contextual crisis tweet summarization using an abstract-extractive method | Twitter |
| **[1]** | NN-based model for warning | Facebook & Twitter |
| **[10]** | ALE based on FFNN to aid in warning & educating | All the Web |
| **[5]** | LSTM-based ALE to aid in education & warning | All the Web |
| **[2]** | Model based on LSTM | All the Web |
| **[7]** | RNN-based model | Smart Data & the Web |
| **[3], [6], [12]** | CNN-LSTM-based Using Crisis Management to Improve Education and Warning | Smart Data & All the Web |
| **Our new Approach** | CNN-LSTM-based Hybrid Model for Agricultural Environment monitoring via Environmental Education and Reliable Information, sustainability and inclusiveness | Multimodal DL (Smart data, Sensors (IoT) & All the Web) |

## IV. Our Sustainable Farming Model to reduce irrigation water waste, enhance resource efficiency and fight desert locusts

The smart and sustainable farming model's objective is, not only to alert farmers, people and notably local and regional officials, but also but also to act on reducing the waste of irrigation water and combating the desert locust [8]. The proposed solution leverages soil moisture, temperature, and humidity sensors connected to CNN-LSTM-based multimodal deep learning to automate irrigation based on real-time data.

Examine a grid (MxN) that depicts a dynamic system that has M rows and N columns: P measures (words, distortions) that change over time are present in every cell. X, a tensor in $\mathbf{R}^{\mathbf{PxMxN}}$, where $\mathbf{R}$ is the observed range, represents an observation. A periodic sequence $X_1, X_2, ..., X_t$ will be examined. Given the earlier observations $\mathbf{J}$, the most likely sequence $\mathbf{K}$ is predicted by the following :

$$\hat{Y}_{t+1}, \cdots, \hat{Y}_{t+K} = \mathbf{arg} \max_{\mathbf{X}_{t+1},...,\mathbf{X}_{t+K}}$$
$$p(\mathbf{X}_{t+1}, ..., \mathbf{X}_{t+K} \mid \mathbf{Y}_{t-J+1}, ..., \mathbf{Y}_{t-J+K}) \tag{1}$$

By describing pixels inside a patch as measurements, it is evident that the goal of breaking down 2D observations into non-overlapping, non-tiled patches is to predict a spatiotemporal sequence. The network input is represented by the letter e in the equation.:

$$e = (w_1, ..., w_i, ..., w_n) \tag{2}$$

comprising a limited vocabulary of words $w_i \in \mathbf{W}$ from $\mathbf{V}$. The content set $\mathbf{C^n}$ is derived from social media, smart data and sensors (IoT).

One of the novel concepts of multimodal machine learning is multimodal fusion. Early, deep, late, and hybrid fusion are the many fusion types. This current research suggests a hybrid fusion method by combining the benefits of early fusion, deep fusion, and late fusion with their benefits [21]. In our case, the advantages of two multimodal fusion techniques are combined in hybrid fusion, namely: late fusion for social networks and smart open health data with early fusion for devices (IoT), modelled as the following formula:

$$f_L = (T_L(T_{L-1}(...T_2(T_1(f_0^{M_{IoMT}})))))\oplus(T_L^{M_SD}( \tag{3}$$

$$T_{L-1}^{M_SD}(...T_1^{M_SD}(f_0^{M_SD}))\oplus T_L^{M_SN}(T_{L-1}^{M_SN}(...T_1^{M_SN}(f_0^{M_SN})))$$

At layer l, $f_l$ is the Multimodal Deep Learning's devices (IoT), Smart Data (SD) and Social Networks (SN)) feature mapping.

The $M_{IoMT}, M_{SD}$ & $M_{SN}$ modalities of devices (IoM), Smart Data (SD) and Social Networks (SN) respectively, in the Layer l, are

$$f_l^{M_{IoMT}}, f_l^{M_{SD}} \ \& \ f_l^{M_{SN}}, l \in (1, 2, \ldots, L)$$

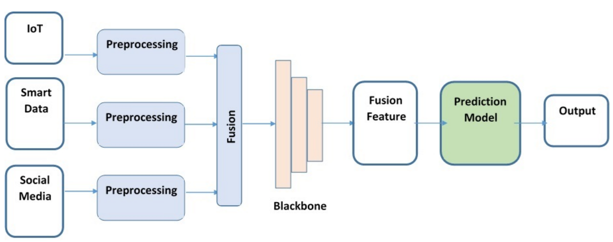


Fig. 2. Example of Hybrid Multimodal Fusion.

Big data mostly refers to information that is too large, too complex, too varied, or changing too quickly for current data processing methods to accurately and quickly analyze and use. Taking into account the quality, security, and protection of the data and their usage, as well as the technological oversight of the data volume, smart data goes beyond this fundamental concept of extensive data and refers to the high-quality and meaningful insights that may be obtained from vast heterogeneous datasets. As a result, the data generates knowledge. To provide comprehensive, compiled, and useful business insights, Smart Data continuously combines the analysis of this data with human intelligence.

Data (signals) from sensors, online content (text, audio, video, and images), images of plants (dry or wet) or soil (dry or wet), social media interactions, information supplied by smart devices (such as smartphones, smart meters, and electronic tags), geolocation data, and more are all included in big data [22].

The major technological difficulty of keeping enormous amounts of data from several channels on massive hard drives that are readily accessible from anywhere in the world is addressed by big data. In the event of an incident, data is always recovered and kept safe.

The next step in optimizing the error function is to "find the coefficient vector $\mathbf{w}^*$ minimizing the error." It is a convex mistake. After determining the ideal $\mathbf{w}^*$ vector, we get the classifier: To determine their probability of failure, use any independent sample that is available. Every neuron in a CNN is linked to the layer below it. They are susceptible to over-learning, nevertheless, because of their complete connection.

$$\forall n \in [1, 2n_c^{[l]}] \qquad Conv(a^{[l-1]}, K_{x,y}^{(n)}) =$$

$$\phi^{[l]}(\sum_{i=1}^{n_H^{[l-1]}} \sum_{j=1}^{n_W^{[l-1]}} \sum_{k=1}^{n_C^{[l-1]}} K_{i,j,k}^{(a)} * a_{x+i-1,y+j-1,k}^{[l-1]} * b_n^{[l]}) \tag{4}$$

$$Dim(Conv(a^{[l-1]}, K^{(n)}) = (n_H^{[l]}, n_W^{[l]})$$

CNNs learn the handcrafted filters in traditional algorithms with minimal to no preprocessing. The effective RNN design for sequence learning, the LSTM, introduces the Memory Cell, an arithmetic unit that takes the place of hidden layer neurons. Information is stored in LSTMs. The RNN degradation issue has been resolved. Both LSTM. Unknown time series can be processed, classified, and predicted by LSTMs [2]. The model is trained by backpropagation. Figure 2 illustrates how the hybrid model based on Deep CNN-LSTM functions.

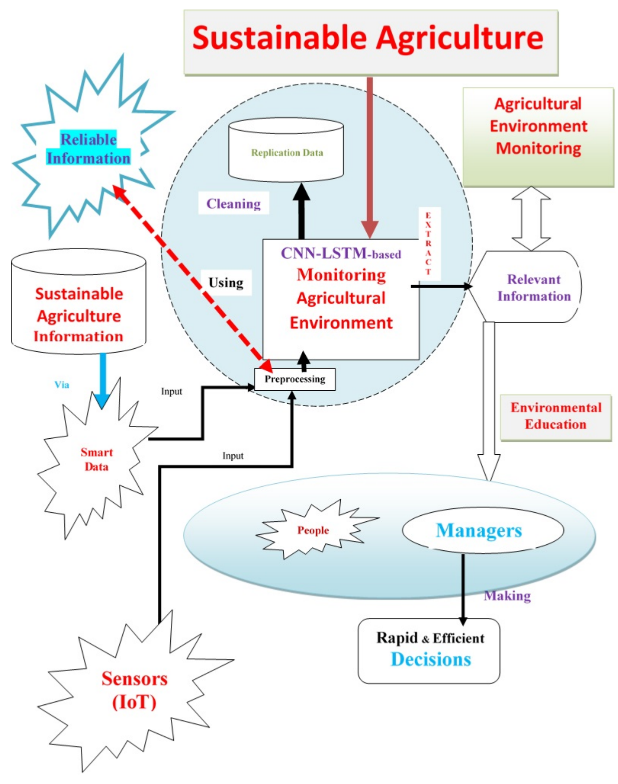


Fig. 3. The Deep CNN-LSTM-based hybrid model's operation

Through neural learning, the conversion of $e_i$ into $e_k$ is defined by:

$$\exists j \in [1, M] \quad | \quad h_j \in \mathbf{H} \quad \& \quad \exists l \in [1, L] \quad | \quad w_l \in \mathbf{W}/$$

$$\max_{k\in[1,K]} \left\{ e_i \longrightarrow e_k = \{e_i \,/\, e_i \; Relevant\; for\; (h_j\;,\; w_l) \right\} \quad (5)$$

$$with \quad i \;\in\; [1, N] \quad \& \quad e_i \;\notin\; [\mathbf{R} + \mathbf{D} + \mathbf{F}]$$

where

$\mathbf{H}$ denotes the $M$ Hashtags set, $\mathbf{W}$ the $L$ keywords set, $\mathbf{F}$ the false alerts set, $\mathbf{R}$ the re-tweets set, and $\mathbf{D}$ the duplicate contents set.

The study's objective is to maximize $\mathbf{K}$, the set size $\mathbf{E_K}$. The output $y_t$ is computed once all of the combined results have been transmitted back and forth using **sigmoid function**. Text and images are now preprocessed before being sent to the extended BERT for embedding. Bert combines a visual method with his ability to depict the link between text and image. A CNN codes each image region, and Bert's model is coupled with a region-based visual feature extractor. The entering text is connected to these graphic elements. Bert's model is fed by this.

Pre-feature extraction: Bert's layer is sufficiently robust. Bert's capacity to pick up semantic features is enhanced with a pre-feature extractor: Convolutional Transform (PCT) in Position, and This pre-feature extractor is formed by Multi-head Self-Attention (MSA).

Concatenation feature: Text and picture latent features are combined to get the necessary feature representations.

Including error functions, learning data collecting, pre and post-data processing, various activation functions, starting weights, and learning algorithms, this research study offers a methodical modeling approach to smart and sustainable farming monitoring. Finding the optimal architecture is the focus of this study, even if performance is influenced by a number of factors.

This has motivated developing a smart irrigation system using IoT and advanced technologies to enable efficient managing water, fighting desert locusts, and supporting sustainable agriculture in Africa (especially in Algeria). This main study objective is to enhance managing irrigation and fighting desert locusts by enabling real-time monitoring of climatic conditions, desert locust populations, and crop vital needs [18]–[20]. The central problem addressed is the need to minimize excessive water consumption while monitoring the arrival of the first locusts, maximizing crop yields, and taking into account economic pressures and environmental constraints. Integrating precision agriculture technology into :

- crop,
- greenhouse, and
- livestock management

also enables the remote automation of processes such as

- irrigation systems,
- feeding systems,
- anti-freeze fans,
- barn temperature management systems, and
- many more.

This approach :

- increases productivity,
- reduces waste, and
- helps farms operate more sustainably, making it a cornerstone of modern agricultural practices:
  - **Precision Pest Management:** IoT agricultural devices monitor pest populations and environmental conditions to predict and prevent outbreaks before they harm crops. By deploying targeted pest control measures, farmers minimize chemical use and ensure healthier harvests, supporting both efficiency and environmental health.
  - **Livestock Monitoring:** IoT in agriculture can include wearable devices that track the health, location, and activity levels of livestock in real time, alerting farmers of illness or injury.

### A. Discussion of Results

We monitored and examined a particular pilot farm in the Algerian highlands. Over the past two years, the managers and staff of this farm have made extraordinary efforts to achieve pre-established and pre-set objectives.

Modern Algerian agriculture faces increasing challenges related to the efficient management of water resources [18], [19], the ongoing invasion of desert locusts and the optimization of agricultural productivity, exacerbated by global climate change while integrating advanced technologies: high-performance sensors and IoT could increase agricultural production and minimize economic losses [20]. These objectives consist of improving yields while enhancing all the necessary conditions for smart and sustainable agriculture. These include, for example:

- reducing the waste of precious irrigation water
- combating desert locusts

These two elements appear to be very important in the management of this farm, as the production capacity for milk, vegetable crops, white and red meat has been improved, enough to cover all the summer holidays, the sacrifices for the Eid religious festivals, not to mention the daily needs of the entire local and surrounding (regional) population. The percentage of reduction in irrigation water waste is shown in Table 5, Figure 3 and Figure 4. Deep learning and neural

TABLE IV
AN OVERVIEW OF THE RESULTS OF THE MULTIMODAL SOURCES OF SMART AND SUSTAINABLE FARMING DURING THIS PERIOD.

| Models | RMSE | MAE | $R^2$ |
|---|---|---|---|
| **Naive Bayes (NB) [26]** | 13,912 | 13,333 | 0,177,407,75 |
| **Support Vector Machine (SVM) [11]** | 17,000 | 15,000 | 0,190,357,14 |
| **Neural Network via Twitter& Facebook [1]** | 31,000 | 23,333 | 0.317,500 |
| **Feed-forward Neural Network [10]** | 55,500 | 33,000 | 0.491,578,93 |
| **RNN via Smart Data & All the Web [7]** | 54,000 | 22,666 | 0.458,208,66 |
| **LSTM via Smart Data & the Web [5]** | 56,000 | 30,666 | 0.499,959,51 |
| **BLSTM via Smart Data & the Web [2]** | 72,000 | 41,666 | 0.749,959,51 |
| **Hybrid Deep CNN-LSTM via All the Web [3], [4], [6]** | 83,000 | 51,333 | 0.856,043,96 |
| **Our New Approach** | 87,400 | 53,333 | 0,918,163,42 |

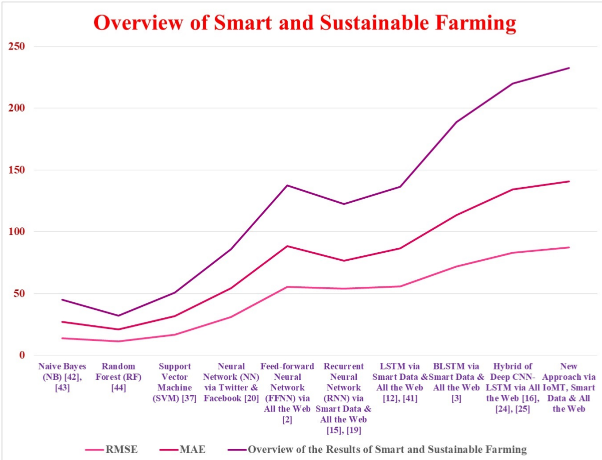


Fig. 4. Overview of the results achieved by smart and sustainable farming during this period..

networks are being used more and more in a variety of applications. Their primary benefit is that, like people, they learn from examples. Their propensity for generalization is their second advantage. This entails recognizing and responding equitably. The datasets were separated into training and verification sets for the investigation. Various deep learning models are trained using these learning sets. The models are tested on the verification set.

You can now examine your hashtags and keywords from your examine menu. The network as a whole is monitored (Listen & Track). Learning is set to Feedforward NN, Recurrent NN, LSTM, BLSTM, CNN, CNN-LSTM, CNN-BLSTM, or Mixed. The administrator can quickly track the remediation process's progress using the View menu. Even cautions and alarms are automatically generated. The user can alter learning, listen and monitor, modify, notify, or warn if they want to step in. With this hybrid technique, various keywords, manually annotated, pertinent, and duplicated content can be shown. In terms of managing content, it permits streaming or a message (particularly for novice users).

## V. Conclusion and outlook

Every facet of sustainable Farming was examined in the research study. The study intends to investigate how agricultural educational establishments contribute to waste of our precious irrigation water in the context of sustainability and inclusivity, examine how global change impacts population susceptibility to a sustainable environment, and assist in the development of strategies that empower communities to better manage our precious irrigation water, fighting desert locusts in Algeria and its effects on sustainability and inclusivity. Every facet of smart and sustainable farming training was examined in the research study. The study intends to investigate how educational agricultural institutions contribute to reduce irrigation water waste and fight desert locusts in Algeria in the context of sustainability and inclusivity, examine how global change impacts population susceptibility to a sustainable environment, and assist in the development of strategies that empower communities to better manage reduce irrigation water waste, fight desert locusts in Algeria and its effects on sustainability and inclusivity. The research provides a CNN-LSTM-based ad hoc real-time hybrid model for smart and sustainable farming monitoring with environmental education. It is predicated on earlier research. A novel multi-view acquisition model from multimodal sources serves as its foundation. For smart and sustainable farming surveillance, this method is quite helpful. Additionally, it warns the neighborhood about possible mishaps such as such as a possible and imminent future invasion of desert locusts. We can assist and receive assistance. Content is written casually, without proper syntax, logical reasoning, noise, spelling mistakes, abbreviations, etc., especially in times of crisis. This may cause the dataset to have a domain bias. Simultaneously, there may be distinct motivations for content in other languages. The hybrid model's attributes were created by examining particular real content reactions. A CNN-LSTM-based ad hoc real-time hybrid model for environmental education and smart and sustainable farming monitoring is presented in the paper. It is predicated on earlier research. A novel multi-view acquisition model from several sources serves as its foundation.

This CNN and LSTM-based hybrid agricultural surveillance model [2] also uses smart data to gather pertinent information for environmental education aimed at alerting irrigation water waste and imminent future invasion of desert locusts. It is predicated on our earlier sophisticated real-time models for handling man-made or natural crashes. This is only the start. In order to warn by teaching, the project plans to incorporate all other stages of agricultural environment observation. It is also intended to employ all deep learning methods and hybrids to enhance the intelligent models. There are several possible future applications for the study. The utilization of several languages (particularly French and possibly Amazigh) and extracting valuable information for the recovery and rescue of escaped animals. The real-time paradigm would be expanded through the use of big data to mine information on past agricultural events and assess potential alerts to control costs.


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## Declarations


*Funding does not require any funding: this work is within the framework of research


* Conflicts of interest: All metrics are the property of the LMA laboratories of the University of Béjaia and LIMPAF of the University of Bouira, Algeria

*Availability of data and material: The data that support the findings of this study are available from the corresponding author but restrictions apply to the availability of these data, which were used in the laboratories LMA from the University of Bejaia in Algeria for the current study, and so are not publicly available. Data are however available from the corresponding author upon reasonable request and with permission of Managers of LMA laboratory.

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